

Supplement — Aggregation-aware monthly CHD/HF thermal associations, Hong Kong, 2013–2023

Companion to [chd_hf_thermal_associations_2013_2023.md](#). All health-effect numbers below are drawn from `outputs/release_chd_hf/` unless labelled SYNTHETIC_CALIBRATION or REAL_PUBLIC_HK0. Provenance for CHD/HF association rows is HA_APPROVED_AGGREGATE.

S0. Reporting checklist

Item	Location
Estimand (monthly count ratio; days offset)	Main Methods; S2
Outcome construction (first recorded hospitalisation after diagnosis; column-label semantics open)	Main Methods; S1
Complete 12-contrast continuity table	<code>outputs/release_chd_hf/tables/table2_core_models.csv</code>
Claim ledger identity CVD-01 to CVD-12	<code>outputs/release_chd_hf/tables/claim_ledger_v2.csv</code>
SE ladder Model/HC1/NW3/NW6	<code>outputs/release_chd_hf/tables/table4_uncertainty_ladder.csv</code> Figure 5
Multiplicity (BH q across 12)	Table 2 <code>q_value_core_bh</code>
Residual ACF / Ljung-Box	S6; <code>outputs/release_chd_hf/supplement/chd_pathway_core_diagnostic_residuals</code> <code>outputs/release_chd_hf/supplement/hf_pathway_core_diagnostic_residuals</code>
Joint vs separate / VIF	S3
Trend, depletion, lag, influence, unoffset INGARCH	S5
Weather source lock + five reproduction checks	S4
M D calibration F1.0 / F1.1 / F1.2	S7
Provenance firewall	S8
Reproducibility commands	S9
Scope exclusions	S10
Authorship / ICD / stroke governance items	authorship_and_governance_decisions.md

S1. Outcome and exposure definitions

S1.1 Cohort and event

- Cohort universe: people diagnosed with type 2 diabetes and/or hypertension during 2013–2023.
- Event: first recorded hospitalisation after the patient's first diagnosis record for CHD or HF.
- Admission cause: not recorded.

- Grain: territory × calendar month, January 2013–December 2023 (132 months).
- Observed columns (gitignored source): CHD chd_inpatient; HF hf_inpatient.
- Care-setting label: inpatient semantics are inferred from column names; confirmation remains human-owned.
- Stroke: named in covering correspondence; file not attached; no stroke estimates.

S1.2 Outcome totals

From outputs/release_chd_hf/tables/table1_outcome_summary.csv:

Outcome	Months	Total events	Monthly mean
CHD	132	156,156	1,183.0
HF	132	29,681	224.9

S1.3 Thermal exposures used in the continuity panel

Internal pathway IDs are retained here for archive mapping only; they do not appear in the journal scientific body.

Pathway ID	Exposure	Contrast
P01A	Monthly mean temperature	per 1 °C
P02A	Monthly mean of daily maximum temperature	per 1 °C
P02B	Monthly mean of daily minimum temperature	per 1 °C
P04A	Official hot nights ($T_{\min} \geq 28$ °C)	per 5 days
P04B	Official cold days ($T_{\min} \leq 12$ °C)	per 5 days
P04C	Official very hot days ($T_{\max} \geq 33$ °C)	per 5 days

S1.4 Denominator

Analysis-of-record offset: $\log(\text{days in month})$. Population aged 35+ × days is an ecological sensitivity only. It is not person-time among members of the diabetes or hypertension cohort still at risk of a first CHD or HF hospitalisation.

S2. Complete model table index

Table / figure	Path	Role
Table 1	outputs/release_chd_hf/tables/table1_outcome_summary.csv	Outcome summary
Table 2	outputs/release_chd_hf/tables/table2_continuity_NW6.csv	Continuity
Table 3	outputs/release_chd_hf/tables/table3_sobarscore_summary.csv	Robustness
Table 4	outputs/release_chd_hf/tables/table4_seasonal_pattern.csv	Seasonal pattern
Claim ledger v2	outputs/release_chd_hf/tables/CVDall_the_CVD_12_with_tiers	CVD all the CVD 12 with tiers
Analysis availability	outputs/release_chd_hf/tables/analysis_blockchain_availability.csv	Blockchain availability
Figure 1	outputs/release_chd_hf/figures/indexed_outcome_series.png	Indexed series
Figure 2	outputs/release_chd_hf/figures/seasonal_pattern.png	Seasonal pattern
Figure 3	outputs/release_chd_hf/figures/forest_forest.png	Forest forest
Figure 4	outputs/release_chd_hf/figures/energy_depletion_and_depletion_sensitivity	Energy depletion and depletion sensitivity
Figure 5	outputs/release_chd_hf/figures/energy_depletion_and_depletion_sensitivity	Energy depletion and depletion sensitivity

S2.1 Pathway panel archives

- outputs/release_chd_hf/supplement/combined_pathway_panel_estimates.csv
- outputs/release_chd_hf/supplement/chd_pathway_panel_forest.png
- outputs/release_chd_hf/supplement/hf_pathway_panel_forest.png
- Core diagnostics: outputs/release_chd_hf/supplement/chd_pathway_core_diagnostics.csv, outputs/release_chd_hf/supplement/hf_pathway_core_diagnostics.csv
- Core fit summary: outputs/release_chd_hf/supplement/cvd_core_model_fit.csv
- Robust SE/offset/family ladder: outputs/release_chd_hf/supplement/cvd_core_robust_estimates.csv

Joint extreme-day and nested heat-month structures remain exploratory. Separate continuity IDs P01A, P02A/B, and P04A-C map to the manuscript continuity panel.

Legacy pollution, humidity, and influenza pathway fits in the broader archive are not adjusted versions of these separate continuity models.

Continuity models used the NB2 parameterisation $\text{Var}(Y | X) = \mu + \mu^2/\theta$; larger θ indicates less negative-binomial overdispersion.

S2.2 Supplementary Table S1 — complete standard-error ladder

Supplementary Table S1. Negative-binomial count ratios under the days-in-month offset. The point estimate is fixed within each outcome-exposure model; only the covariance estimator changes.

Outcome	Exposure	SE method	Count ratio (95% CI)	p
CHD	Mean temperature / 1 °C	Model	0.993 (0.977–1.009)	0.397
CHD	Mean temperature / 1 °C	HC1	0.993 (0.974–1.012)	0.470
CHD	Mean temperature / 1 °C	NW3	0.993 (0.976–1.011)	0.433
CHD	Mean temperature / 1 °C	NW6	0.993 (0.976–1.010)	0.424
CHD	Mean maximum temperature / 1 °C	Model	0.993 (0.979–1.007)	0.334
CHD	Mean maximum temperature / 1 °C	HC1	0.993 (0.976–1.010)	0.408
CHD	Mean maximum temperature / 1 °C	NW3	0.993 (0.978–1.008)	0.345
CHD	Mean maximum temperature / 1 °C	NW6	0.993 (0.979–1.007)	0.321

Outcome	Exposure	SE method	Count ratio (95% CI)	p
CHD	Mean minimum temperature / 1 °C	Model	0.994 (0.979-1.010)	0.463
CHD	Mean minimum temperature / 1 °C	HC1	0.994 (0.976-1.012)	0.528
CHD	Mean minimum temperature / 1 °C	NW3	0.994 (0.977-1.012)	0.517
CHD	Mean minimum temperature / 1 °C	NW6	0.994 (0.977-1.012)	0.509
CHD	Hot nights / 5 days	Model	1.022 (0.995-1.049)	0.112
CHD	Hot nights / 5 days	HC1	1.022 (0.997-1.047)	0.081
CHD	Hot nights / 5 days	NW3	1.022 (1.0003-1.0439)	0.047
CHD	Hot nights / 5 days	NW6	1.022 (1.002-1.042)	0.032
CHD	Cold days / 5 days	Model	0.995 (0.954-1.037)	0.801
CHD	Cold days / 5 days	HC1	0.995 (0.956-1.035)	0.792
CHD	Cold days / 5 days	NW3	0.995 (0.949-1.042)	0.820
CHD	Cold days / 5 days	NW6	0.995 (0.949-1.043)	0.823
CHD	Very hot days / 5 days	Model	0.999 (0.974-1.025)	0.963
CHD	Very hot days / 5 days	HC1	0.999 (0.975-1.025)	0.962
CHD	Very hot days / 5 days	NW3	0.999 (0.973-1.026)	0.964
CHD	Very hot days / 5 days	NW6	0.999 (0.974-1.025)	0.962
HF	Mean temperature / 1 °C	Model	0.974 (0.956-0.993)	0.007
HF	Mean temperature / 1 °C	HC1	0.974 (0.949-1.000)	0.049
HF	Mean temperature / 1 °C	NW3	0.974 (0.949-1.001)	0.055

Outcome	Exposure	SE method	Count ratio (95% CI)	p
HF	Mean temperature / 1 °C	NW6	0.974 (0.947-1.002)	0.069
HF	Mean maximum temperature / 1 °C	Model	0.981 (0.964-0.998)	0.029
HF	Mean maximum temperature / 1 °C	HC1	0.981 (0.958-1.004)	0.111
HF	Mean maximum temperature / 1 °C	NW3	0.981 (0.959-1.004)	0.098
HF	Mean maximum temperature / 1 °C	NW6	0.981 (0.958-1.005)	0.113
HF	Mean minimum temperature / 1 °C	Model	0.973 (0.956-0.991)	0.003
HF	Mean minimum temperature / 1 °C	HC1	0.973 (0.950-0.997)	0.026
HF	Mean minimum temperature / 1 °C	NW3	0.973 (0.948-0.999)	0.038
HF	Mean minimum temperature / 1 °C	NW6	0.973 (0.947-1.000)	0.050
HF	Hot nights / 5 days	Model	1.003 (0.970-1.037)	0.855
HF	Hot nights / 5 days	HC1	1.003 (0.971-1.037)	0.853
HF	Hot nights / 5 days	NW3	1.003 (0.974-1.033)	0.837
HF	Hot nights / 5 days	NW6	1.003 (0.976-1.031)	0.825
HF	Cold days / 5 days	Model	1.073 (1.023-1.125)	0.004
HF	Cold days / 5 days	HC1	1.073 (1.011-1.138)	0.020
HF	Cold days / 5 days	NW3	1.073 (1.007-1.143)	0.031
HF	Cold days / 5 days	NW6	1.073 (1.006-1.144)	0.031

Outcome	Exposure	SE method	Count ratio (95% CI)	p
HF	Very hot days / 5 days	Model	0.995 (0.964–1.027)	0.759
HF	Very hot days / 5 days	HC1	0.995 (0.963–1.028)	0.765
HF	Very hot days / 5 days	NW3	0.995 (0.964–1.027)	0.759
HF	Very hot days / 5 days	NW6	0.995 (0.963–1.028)	0.764

S3. Joint versus separate exposures and collinearity

File: `outputs/release_chd_hf/supplement/cvd_single_vs_joint_estimates.csv` VIF: `outputs/release_chd_hf/supplement/cvd_exposure_vif.csv` Exposure correlations: `outputs/release_chd_hf/supplement/cvd_exposure_correlations.csv`; Figure S1 `outputs/release_chd_hf/supplement/figureS1_exposure_correlation.png`

After month and trend residualisation:

- VIF for mean maximum and minimum temperature (joint temperature model) = 4.66.
- VIF for hot nights and very hot days (joint extreme-day model) $\approx$ 1.96.
- VIF for cold days in the joint extreme-day set $\approx$ 1.00.

Under days-only offset and Newey-West lag-6 intervals, the CHD hot-night count ratio was 1.022 in the separate hot-night model and 1.045 (1.015–1.075) in the joint extreme-day model. The HF cold-day coefficient was about 1.073 in both structures. Joint inflation of the CHD hot-night coefficient is retained as a collinearity warning.

S4. Source-validation audit (weather)

Provenance: REAL_PUBLIC_HK0. Files:

- `outputs/release_chd_hf/supplement/methods_feasibility/weather_definition_source_lock.csv`
- `outputs/release_chd_hf/supplement/methods_feasibility/weather_definition_source_validation.csv`
- `outputs/release_chd_hf/supplement/methods_feasibility/weather_event_morphology_audit.csv`
- `outputs/release_chd_hf/supplement/methods_feasibility/weather_spillover_exposure_build.md`
- Spillover figure: `outputs/release_chd_hf/supplement/methods_feasibility/figure_spillover_event.png`

S4.1 Five source reproduction checks (all pass)

validation_id	expected	observed	source DOI
li_events_1980_2023	57	57	10.1016/j.atmosres.2024.107845
wang_vhd_days_2006_2015	204	204	10.1016/j.scitotenv.2019.07.039
wang_hn_days_2006_2015	230	230	10.1016/j.scitotenv.2019.07.039
wang_vhd_ge5_event_days_2006_2015	48	48	10.1016/j.scitotenv.2019.07.039
wang_hn_ge5_event_days_2006_2015	56	56	10.1016/j.scitotenv.2019.07.039

Release validation recorded `weather_source_validation_all_pass = TRUE` (`outputs/reports/cvd_real_release_validation_all_pass.txt`)

S4.2 Provisional hot-month / cold-month calendar

Provisional HM/CM flags remain unlocked. Calendar figure: outputs/release_chd_hf/supplement/figureS2
Estimates in outputs/release_chd_hf/supplement/chd_hm_cm_panel_estimates.csv and outputs/release_chd_hf/supplement/hf_hm_cm_panel_estimates.csv are not eligible for primary claims until weather reference rules are locked by the weather co-investigator.

S5. Sensitivity ladders

S5.1 Offset and family

File: outputs/release_chd_hf/supplement/cvd_core_robust_estimates.csv. For each continuity-panel exposure: offsets (days only, population $\times$ days, none); families (negative binomial, quasi-Poisson); SE methods (model, HC1, NW3, NW6). Manuscript Table 2 freezes the days-only / negative-binomial / NW6 continuity slice; Table 4 shows the SE ladder without selecting an interval by null exclusion.

S5.2 Trend and first-event depletion

File: outputs/release_chd_hf/supplement/cvd_trend_depletion_sensitivity.csv; summary ranges in outputs/release_chd_hf/tables/table3_robustness_summary.csv; Figure 4.

CHD hot nights: baseline 1.022; scenario range approximately 1.011–1.025. HF cold days: baseline 1.073; scenario range approximately 1.043–1.113; pre-2020 1.113 (1.053–1.176).

S5.3 Lag

File: outputs/release_chd_hf/supplement/cvd_lag_sensitivity.csv.

Outcome	Exposure	Lag 0	Lag 1	Lag 2
CHD	Hot nights /5	1.022 (1.002–1.042)	1.010 (0.979–1.042)	1.016 (0.987–1.045)
HF	Cold days /5	1.073 (1.006–1.144)	1.073 (1.014–1.135)	1.053 (0.985–1.127)

Lag-one-month temperature is a labelled sensitivity. It is not a recovered daily lag curve.

S5.4 Influence

File: outputs/release_chd_hf/supplement/cvd_influence_sensitivity.csv.

Outcome	Pathway ID	Excluded month	RR after exclusion	Max Cook's D
CHD	P04A	2020-02	1.021 (1.001–1.040)	0.110
HF	P04B	2022-02	1.088 (1.034–1.144)	0.085

S5.5 Count-series dependence (unoffset INGARCH)

File: outputs/release_chd_hf/supplement/cvd_count_timeseries_sensitivity.csv.

Negative-binomial INGARCH(1,1) models were fitted **without an offset**. They reduced residual ACF1 (CHD hot-night residual ACF1 ≈ 0.25 ; HF cold-day residual ACF1 ≈ -0.04) while retaining positive CHD hot-night and HF cold-day associations. These fits are unoffset residual-dependence diagnostics. They are not directly comparable with the days-offset continuity estimates and are not a second primary estimand.

S6. Residual diagnostics

Verified diagnostics for the leading days-offset negative-binomial models were:

Outcome × exposure	n	theta	Pearson dispersion	ACF1	Ljung-Box lag-6 p
CHD × hot nights	132	163.0	1.116	0.508	1.17e-8
HF × cold days	132	188.0	1.149	0.146	0.356

Outcome	Continuity residual ACF1 (approx.)	Range across six continuity exposures	Ljung-Box lag 6
CHD	0.51	0.508-0.534	Rejects white noise
HF	0.15	0.131-0.179	Does not reject

Plots: outputs/release_chd_hf/supplement/chd_core_residual_acf_pacf.png, outputs/release_chd_hf/supplement/hf_core_residual_acf_pacf.png. Tables: outputs/release_chd_hf/supplement/hf_pathway_residual_acf.csv.

S7. M|D calibration scenarios and gates

Provenance: SYNTHETIC_CALIBRATION only. Quarantine folder: outputs/release_chd_hf/supplement/methods_synthetic_calibration. These materials evaluate one clean-room implementation under one synthetic design. They are not CHD/HF health findings.

S7.1 Rationale

Monthly sums erase within-month timing. The Basagaña-Ballester aggregated likelihood provides a constrained route from daily exposure to monthly counts ([Basagaña and Ballester 2024, 2026](#)). Admission of real daily coefficients requires the numerical synthetic-calibration thresholds to pass (analysis_plan/final_reanalysis_protocol_2026-08-10.md, Section 8).

S7.2 F1.0 pilot failure

Artifact: outputs/release_chd_hf/supplement/methods_feasibility/md_pilot_f1_0_failure_note.md. Design: 100 replicates; daily nuisance seasonality = one annual sine/cosine pair. CHD-like null cells were acceptable (Type I ≈ 0.00 -0.06; coverage ≈ 0.94 -1.00). HF-like null cells failed under every kernel (Type I ≈ 0.62 -0.75; coverage ≈ 0.25 -0.38). Interpretation: model-design failure from inadequate nuisance seasonality, not evidence of a real HF cold effect.

S7.3 F1.1 correction

Amendment: replace the harmonic with calendar-month indicators, matching the analysis-of-record seasonality, while remaining inside the 20-parameter cap. The failed F1.0 pilot remains auditable. F1.1 then ran 500 replicates. No new real health model was fitted to choose the correction.

S7.4 F1.2 post-run worst-cell decision

Artifact: outputs/release_chd_hf/supplement/methods_feasibility/md_calibration_f1_2_decision_report
Decision: FAIL_METHODS_FEASIBILITY_ONLY. All gates passed: FALSE. F1.2 re-summarised the existing F1.1 500-replicate fits after the run, using worst-cell rules and unchanged numerical thresholds. It did not refit the simulations; admission remained fail.

Gate	Passed	Value	Requirement
Convergence min	Yes	1.000	≥ 0.95
Null Type I range	No	0.048-0.150	within 0.03-0.08
Coverage min	No	0.840	≥ 0.90
Relative bias max	No	32.77	≤ 0.20
False-sign max	No	0.808	≤ 0.10
Hessian condition	Yes	1.82×10^3	$< 1 \times 10^8$
Divergent fits	Yes	0	< 0.05
Stress coverage min	No	0.842	≥ 0.85

Methods-feasibility figures (not health findings):

Paths:

- outputs/release_chd_hf/supplement/methods_feasibility/figure_calibration_type1_by_outcome_h
- outputs/release_chd_hf/supplement/methods_feasibility/figure_calibration_coverage_by_outcome_h
- outputs/release_chd_hf/supplement/methods_feasibility/figure_calibration_gate_passfail.png

S7.5 Real M|D admission rule

Because the F1.2 post-run re-summary failed, no real daily-exposure coefficient is eligible for manuscript Results. Real constrained M|D fitting additionally requires governed analysis panels that are not redistributed in the public repository.

S8. Provenance firewall

Class	Label	May appear on main claim surfaces?
Governed CHD/HF aggregates and model summaries	HA_APPROVED_AGGREGATE	Yes
Public HKO weather validation	REAL_PUBLIC_HKO	Yes, as exposure methods
Synthetic calibration	SYNTHETIC_CALIBRATION	Supplement methods-feasibility only
Daily mortality AF (Jingwen Liu 2020)	Literature	Introduction/Discussion only; not morbidity coefficients

Class	Label	May appear on main claim surfaces?
Modelled excess deaths (Zhenyuan Liu 2026 medRxiv)	Literature / preprint	Companion baseline only
Source HA xlsx / merged panels	Governed; gitignored	Never committed
Private correspondence PDF	Internal provenance	Not cited; no raw addresses or private quotations

Forbidden promotions (from analysis_plan/final_claim_tiers.yml):

- calibration → real finding;
- exploratory → primary without human headline freeze;
- mortality AF → monthly morbidity effect;
- modelled excess death → observed hospitalisation burden;
- general-population offset → cohort incidence rate.

S9. Reproducibility

Identity checks already recorded:

1. Every Table 2 row matches outputs/release_chd_hf/tables/claim_ledger_v2.csv for RR, CI, p, q, SE method, and provenance.
2. BH q-values recomputed with `p.adjust(..., method="BH")` across the twelve NW6 p-values.
3. Release validation: outputs/reports/cvd_real_release_validation.md — 29/29 checks passed on 10 August 2026.
4. Manifest hashes: outputs/release_chd_hf/release_manifest.csv.
5. Index: outputs/release_chd_hf/RELEASE_INDEX.md.

S9.1 Commands for completed public / release / calibration checks

From the repository root:

```
# Public HK0 spillover morphology + five source reproduction checks
Rscript scripts/35_build_spillover_exposures.R
```

```
# Rebuild disclosure-minimised release package from existing analysis tables
Rscript scripts/38_cns_final_release_artifacts.R
```

```
# Recompute Fl.2 worst-cell gates from existing synthetic raw metrics
# (does not refit real health models; requires outputs/calibration_md/md_calibration_raw_metrics)
Rscript scripts/45_recompute_md_calibration_gates.R
```

```
# Release identity / provenance checks (tables, claim ledger, SE ladder, quarantine policy)
Rscript scripts/34_cvd_real_release_checks.R
```

S9.2 Governed full rerun (local files required)

A complete real-data pathway rerun requires governed monthly CHD/HF aggregates and merged analysis panels that are gitignored and not present in the public checkout:

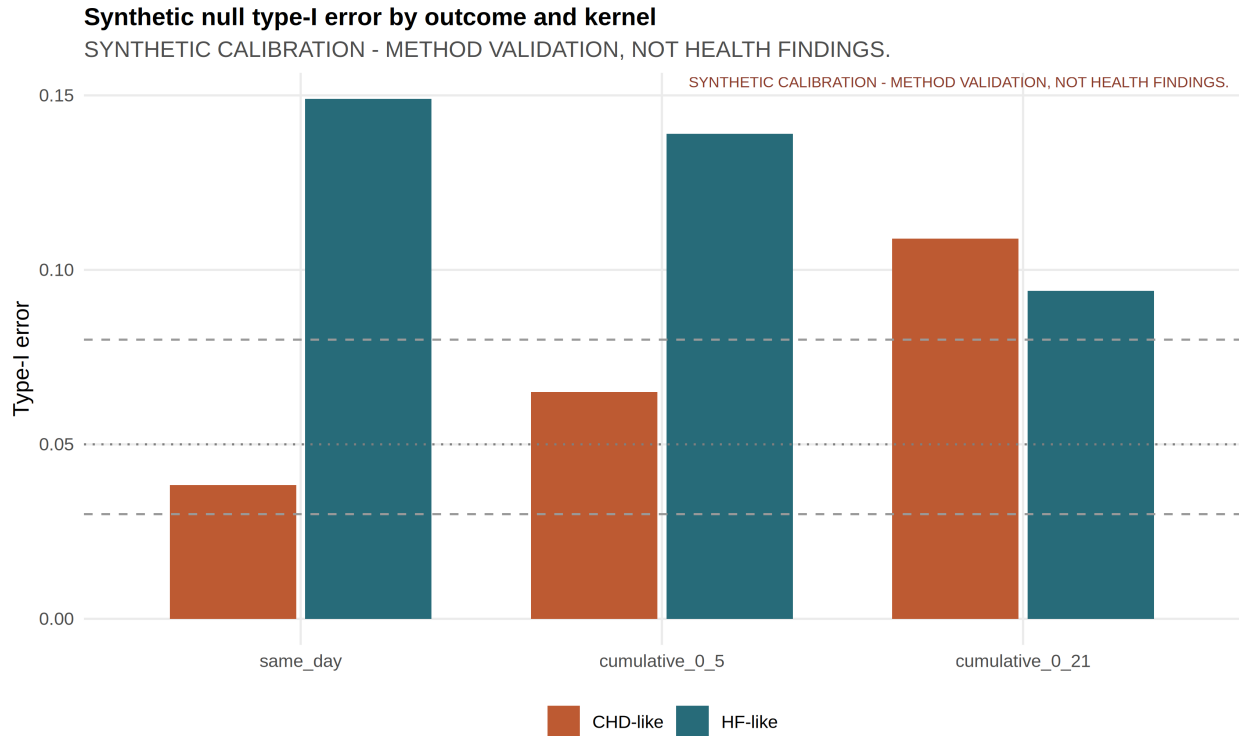


Figure 1: Methods feasibility only — synthetic M|D null Type I by outcome and kernel. Not a CHD/HF health result.

*# Requires local governed inputs under data_raw/ha_secure_placeholder/ (or equivalent)
 # and a constructed analysis panel. Do not commit HA microdata.*

`PATHWAY_MODE=real OUTCOMES=chd,hf Rscript scripts/run_cvd_full_analysis.R`

Without those local governed files, treat release tables under `outputs/release_chd_hf/` as the analysis-of-record artifact set for manuscript drafting.

S10. Scope exclusions

Excluded from the journal scientific claims by design or delivery:

- Stroke results (file not delivered);
- Age/sex or medication effect modification (strata not delivered);
- Principal-diagnosis CHD/HF or AMI claims (admission cause absent);
- Locked hot-month / cold-month primary estimates (weather reference rules unlocked);
- Cohort incidence with the true still-at-risk denominator (denominator unavailable);
- Real daily-exposure M|D coefficients (calibration failed);
- Confounding-adjusted continuity estimates for pollution, humidity, or influenza (legacy/joint archive paths only).

Cloud-runtime availability of governed panels is an internal operations fact and is documented in the supervisor integrated report, not as a journal result.

S11. What this supplement does not contain

- Source monthly HA count files
- Merged health-environment panels with event counts

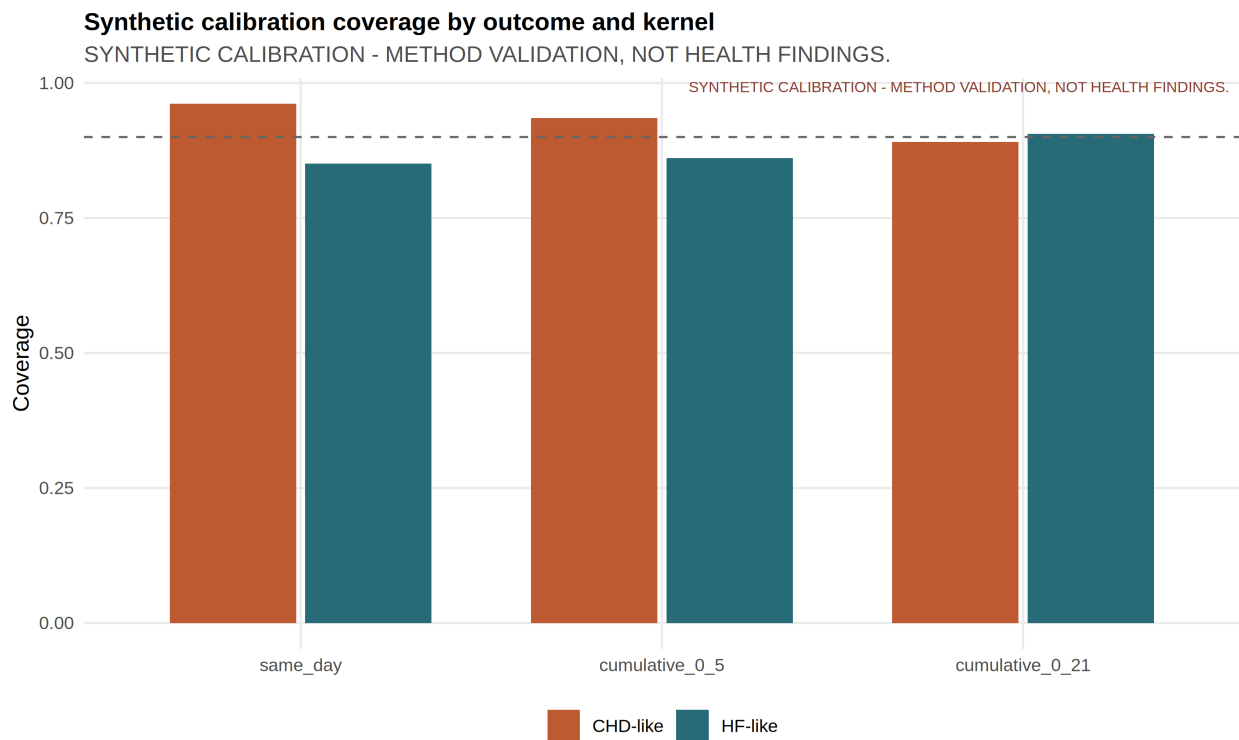


Figure 2: Methods feasibility only — synthetic M|D coverage by outcome and kernel. Not a CHD/HF health result.

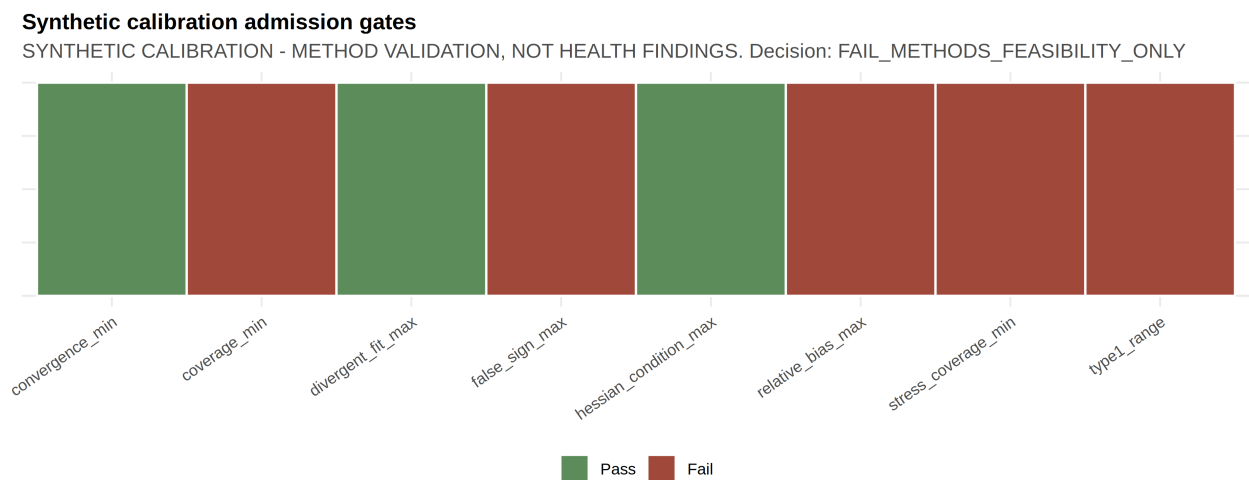


Figure 3: Methods feasibility only — F1.2 gate pass/fail summary. Not a CHD/HF health result.

- Synthetic coefficients presented as health findings
- Private email text or raw addresses
- Final author order or unrecorded headline approval

References

- Basagaña, Xavier, and Joan Ballester. 2024. “Unbiased Temperature-Related Mortality Estimates Using Weekly and Monthly Health Data: A New Method for Environmental Epidemiology and Climate Impact Studies.” *The Lancet Planetary Health* 8 (10): e766–77. [https://doi.org/10.1016/S2542-5196\(24\)00212-2](https://doi.org/10.1016/S2542-5196(24)00212-2).
- . 2026. “Unbiased Estimates Using Temporally Aggregated Outcome Data in Time Series Analysis: Generalization to Different Outcomes, Exposures, and Types of Aggregation.” *Epidemiology* 37 (1): 16–20. <https://doi.org/10.1097/EDE.0000000000001923>.